ENME711 **Student ID:____________________________**

BACHELOR OF ENGINEERING (Hons) – AK3751

Mid-Semester Test

Semester 2, 2019

FLUIDS & THERMODYNAMICS
(Module – Thermodynamics A)

TIME ALLOWED:	1 hour and 15 minutes
MARKS:	TOTAL: **37 marks**
INSTRUCTIONS:	Answer **ALL FOUR questions.**
	All answers except diagrams and sketches ***must be in ink.***
	Write your answers in this answer booklet.
	Correcting fluid is not permitted.
	Exam conditions apply!
NOTES:	Marks allocated to each question are shown in brackets beside each question.
	Candidates should show ***all working*** in calculations.
	Useful Formulae can be found at the end of this exam paper.
	Tables of Thermodynamic Data are supplied.

Question 1 – (5 marks)

Figure 1 shows an unlabelled p-v diagram for a pure substance. Fill in the following information on the diagram:

a) Phase regions:
   i) Liquid
   ii) Solid
   iii) Vapour
   iv) Liquid + Solid
   v) Solid + Vapour
   vi) Liquid + Vapour.

b) Important features:
   i) Critical point
   ii) Triple Line.

c) Process Lines: (clearly identify/label each process)
   i) A constant pressure sublimation process
   ii) A constant temperature boiling process. (from compressed liquid to superheated vapour)

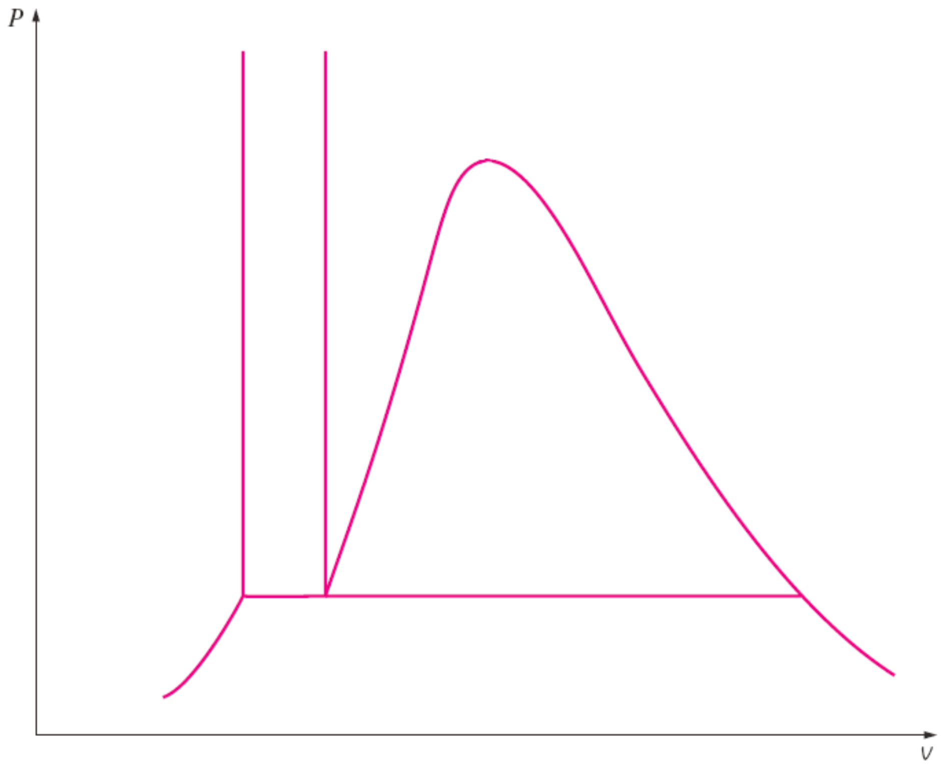

Figure 1

Question 2 – (9 marks)

Fill in the missing information in the following table for ***water-steam***.

	T **(°C)**	***p*** **(kPa)**	***u*** **(kJ/kg)**	***x***	**Phase description**
a.	145		2000		
b.		800	209.34		
c.	270				saturated vapour
d.		30		0.4	
e.		1600	3120.1		
f.	85			0.0	

Question 3 – (10 marks)

A piston cylinder device contains 2 kg of saturated refrigerant (R-134a) liquid and 10 kg of saturated R-134a vapour at a pressure of 200 kPa (Figure 2). Now 250 kJ of heat is transferred to the refrigerant at constant pressure while a 230 V source supplies current to a resistance element within the cylinder for a period of 6 minutes.

a) Determine the current supplied if the final temperature is 70°C

b) Show the process on a *T-v* diagram (with respect to saturation lines).

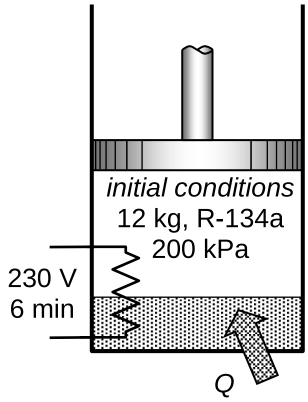

Figure 2

Question 3 (cont'd)

Question 3 (cont'd)

Question 3 (cont'd)

Question 4 – (13 marks)

A piston-cylinder device contains 0.2 kg of air initially at 1.5 MPa and 200°C. The air is first expanded at constant pressure until the volume has doubled. Then, the air is cooled down in a constant-volume process, before it is compressed polytropically to the initial state with a polytopic index of 1.2.

a) Determine the boundary work for each process and the net work of the cycle.
b) Sketch the cycle in a *p*-*V* diagram.

(*Hint:* Polytropic boundary work $W_{12} = \frac{p_2V_2 - p_1V_1}{1-n}$)

Question 4 (cont'd)

Question 4 (cont'd)

Question 4 (cont'd)

Useful Formulae

General & Properties

$$h = u + Pv \qquad y_x = y_\mathrm{f} + x y_\mathrm{fg} \qquad x = \frac{m_\text{sat.vapour}}{m_\text{sat.vapour} + m_\text{sat.liquid}}$$

$$P = \rho g z \qquad v = \frac{V}{m} \qquad \dot{W}_e = VI$$

Ideal Gas

$$PV = mRT \qquad PV = NR_\mathrm{u}T \qquad R_\mathrm{u} = 8.314\,\frac{\mathrm{kJ}}{\mathrm{kmol\ K}} \qquad R = \frac{R_\mathrm{u}}{M}$$

$$\Delta u = C_\mathrm{v}\Delta T \qquad \Delta h = C_\mathrm{p}\Delta T \qquad C_\mathrm{p} = C_\mathrm{v} + R \qquad \gamma = k = \frac{C_\mathrm{p}}{C_\mathrm{v}}$$

First Law – Closed Systems

$$\underbrace{E_\text{in} - E_\text{out}}_{\substack{\text{net energy transfer}\\ \text{by heat and work}}} = \underbrace{\Delta E_\text{system}}_{\substack{\text{change in internal,kinetic}\\ \text{potential,etc energies}}} \qquad W_{b,out} = \int_1^2 P\,dV$$

$$Q - W = \Delta U = m\Delta u \qquad Q_\text{net,in} = W_\text{net,out} \quad \text{(closed cycle)}$$

First Law – Open Systems

$$\dot{m} = \rho \mathcal{v} A = \rho \dot{V} = \frac{\dot{V}}{v} \qquad \mathcal{v} = \text{velocity}$$

Steady Flow Processes

$$\sum \dot{m}_\text{in} = \sum \dot{m}_\text{out} \qquad \dot{Q} - \dot{W} = \underbrace{\sum\nolimits_\text{out} \dot{m}\left(h + \frac{\mathcal{v}^2}{2} + gz\right)}_{\text{for each outlet}} - \underbrace{\sum\nolimits_\text{in} \dot{m}\left(h + \frac{\mathcal{v}^2}{2} + gz\right)}_{\text{for each inlet}}$$

$$\dot{Q} - \dot{W} = \dot{m}\left[(h_2 - h_1) + \frac{\mathcal{v}_2^2 - \mathcal{v}_1^2}{2} + g(z_2 - z_1)\right]$$

Uniform Flow Processes

$$\sum m_\text{in} - \sum m_\text{out} = (m_\text{final} - m_\text{initial})_\text{system} = (m_2 - m_1)_\text{system}$$

$$Q - W = \sum m_\text{out}h_\text{out} - \sum m_\text{in}h_\text{in} + (m_2u_2 - m_1u_1)_\text{system}$$

Question 1 – (5 marks)

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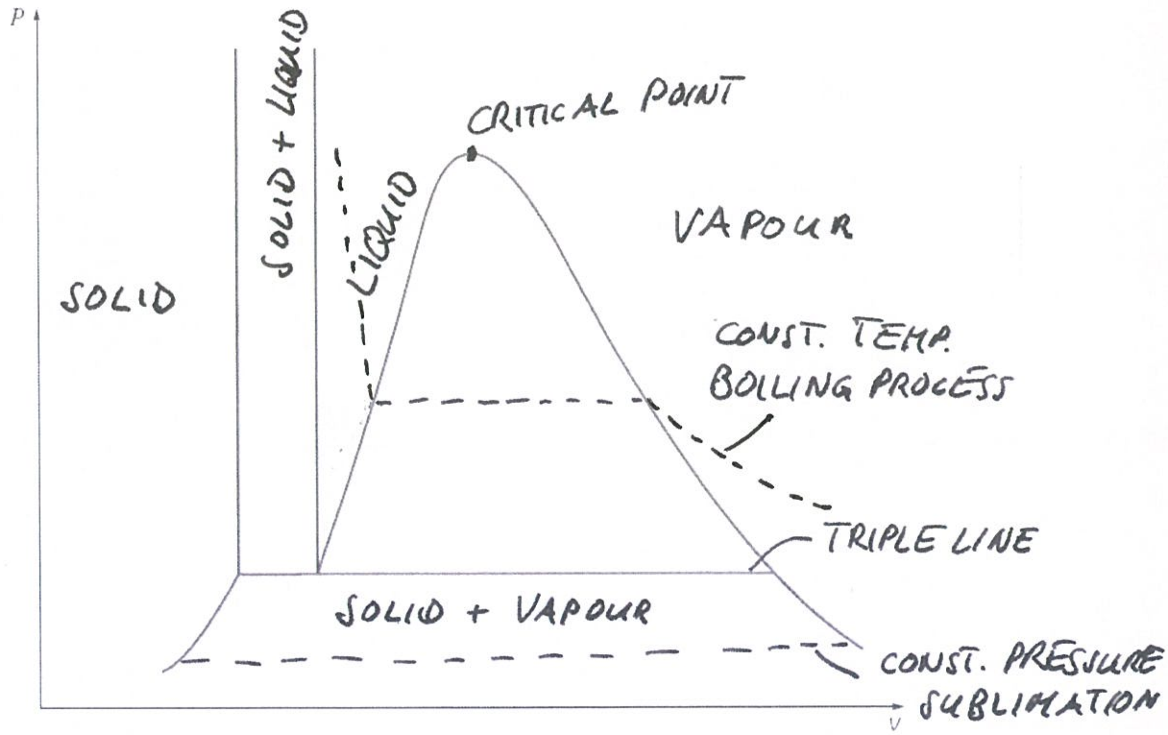


Figure 1

½ each

Σ5

Question 2 – (9 marks)

Fill in the missing information in the following table for ***water-steam***.

	T (°C)	*p* (kPa)	*u* (kJ/kg)	*x*	Phase description
a.	145	415.68	2000	0.72	sat. mixture
b.	50	800	209.34	–	comp. liquid
c.	270	5503.0	2593.7	1.0	saturated vapour
d.	69.09	30	1160.64	0.4	sat. mixture
e.	500	1600	3120.1	–	superheated vapour
f.	85	57.868	355.36	0.0	saturated liquid

½ each

a) $u_f < u < u_g \Rightarrow$ sat. mixture ; $p = p_{sat} = 415.68$ kPa

$$x = \frac{u - u_f}{u_{fg}} = \frac{2000 - 610.13}{1944.2} = 0.715$$

b) $u < u_f \Rightarrow$ compressed liquid ; need to find T where $u = 209.34 \Rightarrow 50°C$

c) sat. vapour $\Rightarrow x = 1$ and $u = u_g$; $p = p_{sat}$ @ 270°C

d) $x = 0.4 \Rightarrow$ sat. mixture ; $T = T_{sat}$ @ 30 kPa

$u = u_f + x u_{fg} = 289.24 + 0.4 \times 2178.5 = 1160.64$

e) $u > u_g \Rightarrow$ superheated vapour ;
$T = 500°C$ @ $u = 3120.1$

f) $x = 0.0 \Rightarrow$ saturated liquid ; $u = u_f$; $p = p_{sat}$ @ 85°C

$\Sigma 9$

Question 3 – (10 marks)

A piston cylinder device contains 2 kg of saturated refrigerant (R-134a) liquid and 10 kg of saturated R-134a vapour at a pressure of 200 kPa (Figure 2). Now 250 kJ of heat is transferred to the refrigerant at constant pressure while a 230 V source supplies current to a resistance element within the cylinder for a period of 6 minutes.

a) Determine the current supplied if the final temperature is 70°C

b) Show the process on a *T-v* diagram (with respect to saturation lines).

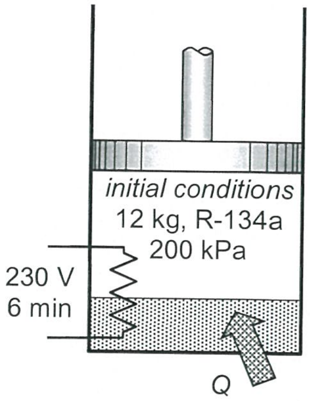
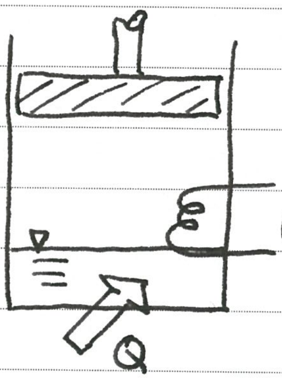


Figure 2



initial condition:

sat. mixture: $x = \frac{10}{12} = 0.833$

$P_1 = 200$ kPa

$Q - W = \Delta U$

$Q - W_e - W_b = \Delta U$

$Q - W_e = \Delta U + W_b = \Delta H$ (2)

initial: $h_1 = h_f + x\, h_{fg} = 38.41 + 0.833 \times 206.09 =$
$= 210.15$ kJ/kg (1)

final: $P_2 = 200$ kPa ; $T_2 = 70°C$ => super heated
from table => $h_2 = 315.03$ kJ/kg (1)

$W_e = Q - m(h_2 - h_1) =$

$= 250 \text{ kJ} - (12 \text{ kg})(315.03 - 210.15) \frac{\text{kJ}}{\text{kg}} =$

$= -1008.56$ kJ (2) (neg. sign => work in)

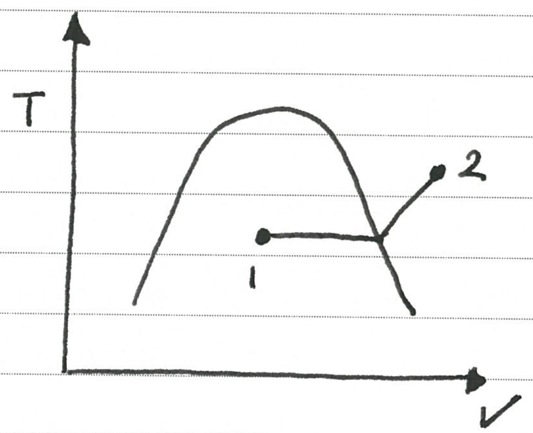
Question 3 (cont'd)

$$W_e = V I \Delta t \Rightarrow \underline{\underline{I}} = \frac{W_e}{V \Delta t} = \frac{1008.56 \times 10^3 J}{(230\,V)(60 \times 6)s} =$$

$$= \underline{\underline{12.18\ A}}$$

2

b)



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$\Sigma 10$

Question 4 – (1~~2~~3 marks)

A piston-cylinder device contains 0.2 kg of air initially at 1.5 MPa and 200°C. The air is first expanded at constant pressure until the volume has doubled. Then, the air is cooled down in a constant-volume process, before it is compressed polytropically to the initial state with a polytopic index of 1.2.

a) Determine the boundary work for each process and the net work of the cycle.

b) Sketch the cycle in a *p*-*V* diagram.

a)

State 1:

$$P_1 = 1.5\ MPa = 1500\ kPa$$
$$T_1 = 200°C = 473\ K$$
$$V_1 = \frac{mRT_1}{P_1} = \frac{(0.2\ kg)(287\ \frac{kJ}{kg\,K})(473\ K)}{(1500\ kPa)} =$$
$$= 0.0181\ m^3 \quad 2$$

State 2:

$$P_2 = P_1 = 1500\ kPa$$
$$V_2 = 2 \times V_1 = 0.0362\ m^3$$

$$W_{12} = \int p\,dV = P_1(V_2 - V_1) = (1500\ kPa)(0.0362 - 0.0181)\ m^3 =$$
$$= 27.2\ kJ \quad 2$$

State 3:

$$V_3 = V_2 \Rightarrow W_{23} = 0 \quad 1$$

$$\frac{P_3}{P_1} = \left(\frac{V_1}{V_3}\right)^n \Rightarrow P_3 = P_1\left(\frac{V_1}{V_3}\right)^n = (1500\ kPa)\left(\frac{1}{2}\right)^{1.2} =$$
$$= 652.9\ kPa \quad 2$$

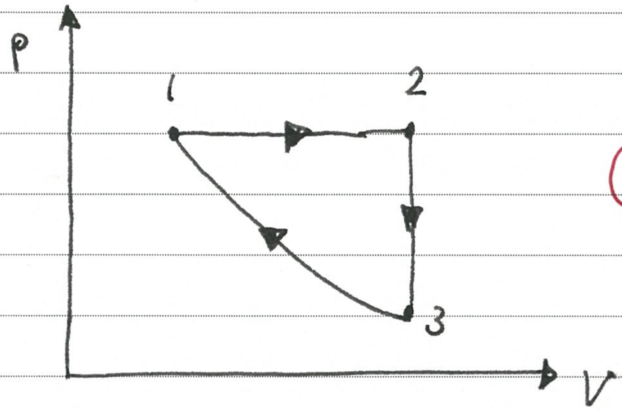
$$W_{31} = \frac{P_1V_1 - P_3V_3}{1-n} = \frac{(1500\ kPa)(0.0181\ m^3) - (652.9\ kPa)(0.0362\ m^3)}{1-1.2} =$$
$$= -17.6\ kJ \quad 2$$

Question 4 (cont'd)

$$W_{net} = W_{12} + W_{23} + W_{31} = 27.2\ kJ - 17.6\ kJ = 9.6\ kJ$$

2

b)



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$\sum 13$